

Diophantine Confinement of the Branch Locus in Syracuse Dynamics

By JOHN A. JANIK

Abstract

We prove that if the transverse walk associated to a Collatz trajectory has eventually linear drift, then the trajectory is eventually bounded and periodic. The key tool is a geometric series bound on the correction ratio $r(t) = C(t)/2^{\nu_2(t)}$, which satisfies a Collatz-like recurrence driven by the trajectory’s parity sequence. We show that the parity sequence of every Collatz orbit lies in the golden mean shift (no two consecutive odd steps), implying $p_{\text{odd}} \leq 1/\varphi^2 \approx 0.382$, within 0.005 of the equilibrium threshold $p_{\text{eq}} \approx 0.387$. Combined with cycle elimination—proved here for all cycle lengths, delegating longer cycles ($\Delta_3 \geq 2$) to Baker–Steiner estimates—this yields a conditional proof of the Collatz conjecture: linear walk drift implies `collatzReaches( $n$ )`. We also prove the converse: reaching 1 implies the K-bound, so the sole critical-path axiom is formally equivalent to the conjecture itself. The argument is formalized in approximately 2800 lines of Lean 4 with seven explicit `sorry` axioms, all identified and categorized.

1. Introduction

1.1. *The conjecture.*

Definition 1.1. The *Collatz map* $T: \mathbb{N} \rightarrow \mathbb{N}$ is defined by

$$T(n) = \begin{cases} n/2 & \text{if } n \equiv 0 \pmod{2}, \\ 3n + 1 & \text{if } n \equiv 1 \pmod{2}. \end{cases}$$

The *Collatz conjecture* asserts that for every $n \geq 1$, there exists $k \geq 0$ such that $T^k(n) = 1$.

Received by the editors February 18, 2026.

2020 *Mathematics Subject Classification.* Primary 11B83; Secondary 37A45, 37B10.

Keywords: Collatz conjecture, correction ratio, golden mean shift, subshift of finite type, cycle elimination, Lean 4 formalization

© XXXX Department of Mathematics, Princeton University.

The conjecture has been verified computationally for all $n \leq 10^{20}$ [6] and remains one of the most resistant open problems in elementary number theory. Surveys of the extensive literature can be found in Lagarias [4, 5].

1.2. *Prior work.* Terras [10] introduced the stopping time and proved that almost all integers (in the sense of natural density) have a finite stopping time. Lagarias [4] formalized the probabilistic heuristic that treats the Collatz iteration as a random walk with drift $\log_2(4/3)$ per odd step. Wirsching [11] developed the ergodic-theoretic perspective, studying transfer operators associated to the dynamics.

Steiner [8] and Simons–de Weger [7] used Baker-type bounds on linear forms in logarithms to establish lower bounds on the period of any nontrivial Collatz cycle. Eliahou [3] showed that no nontrivial cycle has period less than 17087915. Conway [2] proved that Collatz-type problems can be algorithmically undecidable in general, though this does not apply to the specific $3n + 1$ function.

The strongest density result is due to Tao [9], who proved that almost all orbits (in a logarithmic density sense) attain almost bounded values. His approach uses entropy-based methods on the Syracuse random variables and does not yield the full conjecture.

1.3. *Our approach.* The present paper introduces a different mechanism. The argument proceeds in three steps:

- (1) The parity sequence of every Collatz trajectory lies in the *golden mean shift* (the subshift of finite type forbidding the bigram “11”). This constrains the odd-step proportion to $p_{\text{odd}} \leq 1/\varphi^2 \approx 0.382$, just below the equilibrium threshold $p_{\text{eq}} = 1/(1 + \log_2 3) \approx 0.387$.
- (2) The *correction ratio* $r(t) = C(t)/2^{\nu_2(t)}$, which captures the cumulative effect of the “+1” in $3n + 1$, satisfies a geometric series bound when the transverse walk has linear drift. This forces the trajectory to be eventually bounded.
- (3) Cycle elimination—proved directly for cycles with $\Delta_3 = 1$ odd step per period, and following from Baker’s theorem for $\Delta_3 \geq 2$ —shows the only positive cycle is $\{1, 2, 4\}$.

1.4. *Main results.* We state two main theorems. Theorem A is unconditional given the drift hypothesis; Theorem B is a conditional proof of the full conjecture.

THEOREM A (Walk drift implies reaches 1). *Let $n \geq 1$. If the transverse walk $w(t)$ has eventually linear drift with rate $\varepsilon > 0$, and the only positive Collatz cycle is $\{1, 2, 4\}$, then $T^k(n) = 1$ for some $k \geq 0$.*

THEOREM B (Conditional Collatz conjecture). *Assume:*

(H1) *For every $n \geq 1$, there exist $\varepsilon > 0$, T_0 , and K such that $\nu_3(t)/t \leq p_{\text{eq}} - \varepsilon$ and $3\nu_3(t) \leq t + K$ for all $t \geq T_0$.*

(H2) *Baker's theorem on linear forms in logarithms holds for $\alpha_1 = 2$, $\alpha_2 = 3$.*

Then for every $n \geq 1$, there exists $k \geq 0$ such that $T^k(n) = 1$.

Hypothesis (H1) is a quantitative version of the statement that the transverse walk has positive linear drift. The golden mean shift constraint makes this plausible: the gap between $1/\varphi^2 \approx 0.382$ and $p_{\text{eq}} \approx 0.387$ is only 0.005. What remains is to promote this topological bound to a pointwise ergodic statement for individual trajectories—essentially, an ergodic theorem for the Collatz transfer operator, or a spectral gap argument. Hypothesis (H2) is established mathematics [1]; the gap is its formalization in Lean 4.

1.5. *Outline.* Section 2 introduces the winding numbers, transverse walk, and the multiplicative identity that relates trajectory values to their parity counts. Section 3 establishes that every Collatz parity sequence lies in the golden mean shift. Section 4 contains the core result: the correction ratio bound. Section 5 treats cycle elimination. Section 6 assembles the main theorems. Section 7 describes the Lean 4 formalization. Section 8 summarizes computational evidence.

2. Winding numbers and the multiplicative identity

Definition 2.1. For $n \geq 1$, define the *Collatz sequence* by $a_0 = n$ and $a_{t+1} = T(a_t)$. Let $\nu_2(t)$ and $\nu_3(t)$ count the even and odd steps among $0, 1, \dots, t-1$, so that $\nu_2(t) + \nu_3(t) = t$. The *transverse walk* is

$$(2.1) \quad w(t) = \nu_2(t) - \log_2 3 \cdot \nu_3(t).$$

The quantity $w(t)$ measures the excess of even steps over the equilibrium rate $\log_2 3$ per odd step. If $w(t) \rightarrow +\infty$, then even steps dominate sufficiently to force the trajectory toward small values.

PROPOSITION 2.2 (Multiplicative identity). *For all $n \geq 1$ and $t \geq 0$,*

$$(2.2) \quad a_t \cdot 2^{\nu_2(t)} = n \cdot 3^{\nu_3(t)} + C(t),$$

where the correction term $C(t)$ satisfies $C(0) = 0$ and

$$(2.3) \quad C(t+1) = \begin{cases} C(t) & \text{if step } t \text{ is even,} \\ 3C(t) + 2^{\nu_2(t)} & \text{if step } t \text{ is odd.} \end{cases}$$

Proof. By induction on t . The base case $t = 0$ gives $a_0 \cdot 2^0 = n \cdot 3^0 + 0$.

1 If step t is even, then $a_{t+1} = a_t/2$, $\nu_2(t+1) = \nu_2(t) + 1$, $\nu_3(t+1) = \nu_3(t)$,
2 and $C(t+1) = C(t)$. The identity at $t+1$ becomes

$$\frac{a_t}{2} \cdot 2^{\nu_2(t)+1} = a_t \cdot 2^{\nu_2(t)} = n \cdot 3^{\nu_3(t)} + C(t).$$

5 If step t is odd, then $a_{t+1} = 3a_t + 1$, $\nu_2(t+1) = \nu_2(t)$, $\nu_3(t+1) = \nu_3(t) + 1$,
6 and $C(t+1) = 3C(t) + 2^{\nu_2(t)}$. The identity at $t+1$ becomes

$$\begin{aligned} (3a_t + 1) \cdot 2^{\nu_2(t)} &= 3 \cdot a_t \cdot 2^{\nu_2(t)} + 2^{\nu_2(t)} \\ &= 3(n \cdot 3^{\nu_3(t)} + C(t)) + 2^{\nu_2(t)} \\ &= n \cdot 3^{\nu_3(t)+1} + 3C(t) + 2^{\nu_2(t)}. \end{aligned} \quad \square$$

12 *Definition 2.3.* The *correction ratio* is

$$r(t) = \frac{C(t)}{2^{\nu_2(t)}}.$$

16 LEMMA 2.4 (Correction ratio recurrence). *The correction ratio satisfies:*

- 17 • *Even step:* $r(t+1) = r(t)/2$.
- 18 • *Odd step:* $r(t+1) = 3r(t) + 1$.

20 *Proof.* At an even step, $\nu_2(t+1) = \nu_2(t) + 1$ and $C(t+1) = C(t)$, so
21 $r(t+1) = C(t)/2^{\nu_2(t)+1} = r(t)/2$.

22 At an odd step, $\nu_2(t+1) = \nu_2(t)$ and $C(t+1) = 3C(t) + 2^{\nu_2(t)}$, so $r(t+1) =$
23 $(3C(t) + 2^{\nu_2(t)})/2^{\nu_2(t)} = 3r(t) + 1$. $\square$

24 PROPOSITION 2.5. *For all $n \geq 1$ and $t \geq 0$,*

$$(2.4) \quad a_t = n \cdot 2^{-w(t)} + r(t).$$

28 *In particular, if $w(t) \rightarrow +\infty$, then $a_t \approx r(t)$ for large t .*

29 *Proof.* Dividing (2.2) by $2^{\nu_2(t)}$ gives $a_t = n \cdot 3^{\nu_3(t)}/2^{\nu_2(t)} + r(t)$. Since
30 $3^{\nu_3}/2^{\nu_2} = 2^{\nu_3 \log_2 3 - \nu_2} = 2^{-w(t)}$, the result follows. $\square$

31

32

33

3. The golden mean shift

34

35

PROPOSITION 3.1. *If n is odd, then $3n + 1$ is even.*

36

37

Proof. If n is odd, then $3n$ is odd, so $3n + 1$ is even. $\square$

38

39

COROLLARY 3.2. *In any Collatz trajectory, no two consecutive steps can both be odd steps.*

40

41

Proof. An odd step is applied when a_t is odd, producing $a_{t+1} = 3a_t + 1$,
42 which is even by Proposition 3.1. Therefore step $t+1$ is an even step. $\square$

Definition 3.3. The *parity sequence* of the Collatz orbit of n is the sequence $\sigma = (\sigma_0, \sigma_1, \sigma_2, \dots) \in \{0, 1\}^{\mathbb{N}}$ where $\sigma_t = a_t \bmod 2$. The *golden mean shift* $X_\varphi \subset \{0, 1\}^{\mathbb{N}}$ is the subshift of finite type defined by the forbidden bigram 11:

$$X_\varphi = \{\sigma \in \{0, 1\}^{\mathbb{N}} : \sigma_t \cdot \sigma_{t+1} = 0 \text{ for all } t \geq 0\}.$$

THEOREM 3.4. *The parity sequence of every Collatz orbit lies in the golden mean shift X_φ .*

Proof. This is a restatement of Corollary 3.2: the bigram $\sigma_t = 1, \sigma_{t+1} = 1$ would require two consecutive odd values, which is impossible. $\square$

COROLLARY 3.5. *For any Collatz trajectory of length t , the odd-step proportion satisfies $\nu_3(t)/t \leq \lceil t/2 \rceil/t$. In particular, $\limsup_{t \rightarrow \infty} \nu_3(t)/t \leq 1/2$.*

More precisely, the topological entropy of the golden mean shift is $\log \varphi$, where $\varphi = (1 + \sqrt{5})/2$ is the golden ratio. The measure of maximal entropy assigns weight $1/\varphi^2 = (3 - \sqrt{5})/2 \approx 0.382$ to the symbol 1. Therefore, under the maximal-entropy measure, $p_{\text{odd}} = 1/\varphi^2$.

PROPOSITION 3.6. $\log_2 3 < \varphi$.

Proof. We show $\log_2 3 < 8/5 < \varphi$. For the first inequality, $3^5 = 243 < 256 = 2^8$, so $5 \log_2 3 < 8$, giving $\log_2 3 < 8/5$. For the second, $(8/5)^2 = 64/25 < 5 = \varphi + 1/\varphi$; more precisely, $\varphi^2 = \varphi + 1 > 8/5 + 1 = 13/5$, and $(8/5)^2 = 64/25 < 65/25 = 13/5$. $\square$

Remark 3.7. The equilibrium threshold for the walk (2.1) is

$$p_{\text{eq}} = \frac{1}{1 + \log_2 3} \approx 0.38685.$$

At this odd-step proportion, the expected walk increment is zero. The golden mean shift bound gives $p_{\text{odd}} \leq 1/\varphi^2 \approx 0.38197$. The gap is $p_{\text{eq}} - 1/\varphi^2 \approx 0.005$, tantalizingly small. This means the SFT constraint *nearly* suffices to prove positive drift. The remaining challenge is to promote the topological bound $1/\varphi^2$ to a pointwise ergodic statement for individual trajectories. Proposition 3.6 ensures that the inequality goes the right way: since $\log_2 3 < \varphi$, the reciprocal satisfies $1/(1 + \log_2 3) > 1/(1 + \varphi) = 1/\varphi^2$, confirming that the golden mean shift maximum $1/\varphi^2$ lies strictly below p_{eq} .

4. The correction ratio bound

This section contains the core new result.

Definition 4.1. We say the walk has *eventually linear drift* with rate $\varepsilon > 0$ if there exists T_0 such that $w(t) \geq \varepsilon t$ for all $t \geq T_0$.

4.1. *Unrolling the correction recurrence.* Let the odd steps occur at times $\tau_1 < \tau_2 < \tau_3 < \dots$. Write $U_j = \nu_2(\tau_j)$ for the cumulative even-step count at the j -th odd step. Since even steps leave C unchanged and each odd step sends $C \mapsto 3C + 2^{U_j}$, unrolling gives:

LEMMA 4.2 (Correction sum). *After the k -th odd step,*

$$(4.1) \quad C(\tau_k + 1) = \sum_{j=1}^k 3^{k-j} \cdot 2^{U_j}.$$

Proof. By induction. At $k = 1$: $C(\tau_1 + 1) = 3 \cdot 0 + 2^{U_1} = 2^{U_1}$. Assuming the formula at k , the $(k+1)$ -th odd step gives $C(\tau_{k+1} + 1) = 3 \cdot C(\tau_k + 1) + 2^{U_{k+1}}$, which expands to $\sum_{j=1}^k 3^{k+1-j} \cdot 2^{U_j} + 2^{U_{k+1}} = \sum_{j=1}^{k+1} 3^{k+1-j} \cdot 2^{U_j}$. $\square$

At times just after the k -th odd step (where $\nu_2 = U_k$ and $\nu_3 = k$), the correction ratio is

$$(4.2) \quad r = \frac{C(\tau_k + 1)}{2^{U_k}} = \sum_{m=0}^{k-1} 3^m \cdot 2^{-(U_k - U_{k-m})} = \sum_{m=0}^{k-1} 3^m \cdot 2^{-D_m},$$

where $D_m = U_k - U_{k-m}$ is the number of even steps between the $(k-m)$ -th and k -th odd steps.

4.2. *The walk-drift gap bound.*

LEMMA 4.3 (Walk-drift gap bound). *If the walk has eventually linear drift with rate $\varepsilon > 0$, then for all sufficiently large k and all $0 \leq m \leq k$ with $\tau_{k-m} \geq T_0$,*

$$(4.3) \quad D_m \geq m \cdot \alpha, \quad \text{where} \quad \alpha = \frac{\log_2 3 + \varepsilon}{1 - \varepsilon} > \log_2 3.$$

Proof. Apply the drift condition at times τ_{k-m} and τ_k :

$$\begin{aligned} w(\tau_k) &= U_k - \log_2 3 \cdot k \geq \varepsilon \tau_k, \\ w(\tau_{k-m}) &= U_{k-m} - \log_2 3 \cdot (k-m) \geq \varepsilon \tau_{k-m}. \end{aligned}$$

Subtracting gives $D_m - \log_2 3 \cdot m \geq \varepsilon(\tau_k - \tau_{k-m})$. Between the $(k-m)$ -th and k -th odd steps, the trajectory takes m odd steps and D_m even steps, so $\tau_k - \tau_{k-m} = D_m + m$. Therefore $D_m - \log_2 3 \cdot m \geq \varepsilon(D_m + m)$, giving $D_m(1 - \varepsilon) \geq m(\log_2 3 + \varepsilon)$, hence $D_m \geq m\alpha$. $\square$

4.3. *The geometric series bound.*

THEOREM 4.4 (Correction ratio bound). *If the walk has eventually linear drift with rate $\varepsilon > 0$, then $r(t)$ is eventually bounded. Specifically, there exists*

K such that for all $k \geq K$,

$$(4.4) \quad r(\tau_k + 1) \leq \frac{1}{1 - \rho} + 1, \quad \text{where } \rho = 3 \cdot 2^{-\alpha} < 1.$$

Since r only decreases at even steps ($r \rightarrow r/2$), the bound holds at all times $t \geq \tau_K + 1$.

Proof. Split the sum (4.2) at the threshold m_0 such that $\tau_{k-m} \geq T_0$ for all $m \leq m_0$. Since there are at most T_0 odd steps before time T_0 , we have $m_0 \geq k - T_0$.

Recent terms ($m \leq m_0$, i.e., $\tau_{k-m} \geq T_0$). By Lemma 4.3, $D_m \geq m\alpha$, so $3^m \cdot 2^{-D_m} \leq 3^m \cdot 2^{-m\alpha} = \rho^m$. Since $\alpha > \log_2 3$, we have $\rho = 3/2^\alpha < 1$, and

$$\sum_{m=0}^{m_0} \rho^m \leq \frac{1}{1 - \rho}.$$

Early terms ($m > m_0$, i.e., $\tau_{k-m} < T_0$). For these terms, $U_{k-m} \leq T_0$, so $D_m = U_k - U_{k-m} \geq U_k - T_0$. From the walk-drift condition, $U_k \geq k\alpha$, giving

$$3^m \cdot 2^{-D_m} \leq 3^m \cdot 2^{-(k\alpha - T_0)} = 2^{T_0} \cdot 3^m \cdot 2^{-k\alpha}.$$

Since $m \leq k$, we have $3^m \leq 3^k$, and $3^m \cdot 2^{-D_m} \leq 2^{T_0} \cdot \rho^k$. There are at most T_0 such terms, contributing at most $T_0 \cdot 2^{T_0} \cdot \rho^k$, which vanishes as $k \rightarrow \infty$.

Combining: for k large enough that $T_0 \cdot 2^{T_0} \cdot \rho^k < 1$,

$$r(\tau_k + 1) \leq \frac{1}{1 - \rho} + 1. \quad \square$$

COROLLARY 4.5 (Trajectory bound). *Under the hypotheses of Theorem 4.4, the Collatz sequence a_t is eventually bounded:*

$$a_t \leq B = \left\lfloor \frac{1}{1 - \rho} \right\rfloor + 2 \quad \text{for all } t \geq T_1,$$

for some T_1 depending on n , ε , and T_0 .

Proof. From Proposition 2.5, $a_t = n \cdot 2^{-w(t)} + r(t)$. Choose T_1 so that both $n \cdot 2^{-w(t)} < 1$ (possible since $w(t) \rightarrow +\infty$) and $r(t) \leq 1/(1 - \rho) + 1$ hold for $t \geq T_1$. Then $a_t < 1/(1 - \rho) + 2$, and since a_t is a positive integer, the result follows. $\square$

COROLLARY 4.6 (Eventual periodicity). *Under the same hypotheses, a_t is eventually periodic: there exist $T_2 \geq T_1$ and $p \geq 1$ such that $a_{t+p} = a_t$ for all $t \geq T_2$.*

Proof. For $t \geq T_1$, the sequence a_t takes values in the finite set $\{1, 2, \dots, B\}$. Since $a_t \leq B$ for all $t \geq T_1$ and the Collatz map is deterministic, some value must repeat. From the first repetition onward, the sequence is periodic. $\square$

Remark 4.7 (Why $C/2^{\nu_2}$, not $C/3^{\nu_3}$). The ratio $C(t)/3^{\nu_3(t)}$ asks how large the correction is relative to the odd-step growth. Since 3^{ν_3} grows more slowly than 2^{ν_2} when the walk diverges, this ratio can be unbounded even when the trajectory is well-behaved. Indeed, once the sequence enters the cycle $1 \rightarrow 4 \rightarrow 2 \rightarrow 1$, each 3-step period contributes 2 even and 1 odd step, and the recurrence sends $C \mapsto 3C + 2^{\nu_2}$ at the odd step. The ratio $C/3^{\nu_3}$ then grows as $(4/3)^m$ per cycle—unbounded.

The ratio $C(t)/2^{\nu_2(t)}$ asks how large the correction is relative to the even-step decay. Walk divergence ensures 2^{ν_2} grows fast enough to absorb the correction.

5. Cycle elimination

It remains to show that any cycle entered by a walk-divergent trajectory must be the trivial cycle $\{1, 2, 4\}$.

5.1. The cycle equation.

PROPOSITION 5.1 (Cycle equation). *Suppose a_t is periodic with period p , and let Δ_2, Δ_3 be the number of even and odd steps per period ($\Delta_2 + \Delta_3 = p$). If $2^{\Delta_2} > 3^{\Delta_3}$ (walk divergence), then the minimum cycle element $c_{\min}$ satisfies*

$$(5.1) \quad c_{\min} = \frac{S}{2^{\Delta_2} - 3^{\Delta_3}},$$

where $S = \sum_{j=1}^{\Delta_3} 3^{\Delta_3-j} \cdot 2^{e_j}$ and e_j is the number of even steps preceding the j -th odd step within one period.

Proof. Apply the multiplicative identity (2.2) at time t (within the periodic regime) and at time $t+p$. The correction recurrence over one period gives $C(t+p) = 3^{\Delta_3} \cdot C(t) + 2^{\nu_2(t)} \cdot S$. Substituting into the identity at $t+p$ and using $a_{t+p} = a_t = c_0$:

$$c_0 \cdot 2^{\nu_2(t)+\Delta_2} = c_0 \cdot 3^{\Delta_3} \cdot 2^{\nu_2(t)} + 2^{\nu_2(t)} \cdot S.$$

Dividing by $2^{\nu_2(t)}$ gives $c_0 \cdot 2^{\Delta_2} = c_0 \cdot 3^{\Delta_3} + S$, hence $c_0 = S/(2^{\Delta_2} - 3^{\Delta_3})$. $\square$

5.2. One-odd-step cycles.

THEOREM 5.2. *The only Collatz cycle with $\Delta_3 = 1$ odd step per period is the trivial cycle $\{1, 2, 4\}$.*

Proof. With $\Delta_3 = 1$, the sum S has a single term: $S = 2^{e_1}$, where $e_1 \in \{0, 1, \dots, \Delta_2 - 1\}$. The cycle equation gives

$$c_0 = \frac{2^{e_1}}{2^{\Delta_2} - 3}.$$

For c_0 to be a positive integer, $(2^{\Delta_2} - 3) \mid 2^{e_1}$. Since $2^{\Delta_2} - 3$ is odd for $\Delta_2 \geq 2$, and 2^{e_1} is a power of 2, the only way an odd number divides a power of 2 is if that odd number is 1. Therefore $2^{\Delta_2} - 3 = 1$, giving $\Delta_2 = 2$.

This yields $p = 3$ and $c_0 = 2^{e_1}$ with $e_1 \in \{0, 1\}$. Both values produce the same cycle $\{1, 2, 4\}$ (starting from different points).

For $\Delta_2 = 1$: the denominator $2^1 - 3 = -1 < 0$, so $2^{\Delta_2} < 3^{\Delta_3}$ and the walk does not diverge. This case is excluded by hypothesis. $\square$

5.3. Multi-odd-step cycles and Baker's theorem.

Remark 5.3 (Baker-type bounds for $\Delta_3 \geq 2$). For $\Delta_3 \geq 2$, the cycle equation (5.1) requires the denominator $2^{\Delta_2} - 3^{\Delta_3}$ to divide S exactly. Baker's theorem on linear forms in logarithms [1] gives an effective lower bound:

$$|2^{\Delta_2} - 3^{\Delta_3}| \geq C' \cdot 3^{\Delta_3} / \Delta_2^{1+\delta}$$

for effective constants $C', \delta > 0$. Combined with the upper bound $S < 3^{\Delta_3} \cdot 2^{\Delta_2}$, this forces $c_0 < \Delta_2^{1+\delta} \cdot 2^{\Delta_2} / C'$.

Steiner [8] initiated this approach. Eliahou [3] proved there are no non-trivial cycles with period $p \leq 17\,087\,915$. Simons and de Weger [7] extended these bounds further. We do not reproduce the Baker-type arguments here; we treat (H2) as a hypothesis in Theorem B.

6. Main results

We now assemble the components into the main theorems.

Proof of Theorem A. Let $n \geq 1$ and assume the walk $w(t)$ has eventually linear drift with rate $\varepsilon > 0$.

By Theorem 4.4, the correction ratio $r(t)$ is eventually bounded by $B_r = 1/(1 - \rho) + 1$.

By Corollary 4.5, the trajectory satisfies $a_t \leq B = \lfloor B_r \rfloor + 2$ for all $t \geq T_1$.

By Corollary 4.6, the trajectory is eventually periodic.

By hypothesis, the only positive Collatz cycle is $\{1, 2, 4\}$. Since a_t is eventually periodic and must enter this cycle, we conclude $a_t = 1$ for some t . $\square$

We now address the two hypotheses needed for the full conjecture.

Discussion of (H1). Hypothesis (H1) asserts that for every $n \geq 1$, the odd-step proportion $\nu_3(t)/t$ is eventually bounded below $p_{\text{eq}} = 1/(1 + \log_2 3)$ by a uniform gap $\varepsilon > 0$, and that the linear bound $3\nu_3(t) \leq t + K$ holds for large t . In the formalization, only the K-bound $3\nu_3(t) \leq t + K$ is an axiom (`nu3_linear_bound`); the ε -gap follows since $p_{\text{eq}} > 1/3$ (from $\log_2 3 < 2$, i.e., $3 < 4$). The converse also holds: if $T^k(n) = 1$ for some k , then the trajectory enters the cycle $1 \rightarrow 4 \rightarrow 2 \rightarrow 1$, which contributes exactly one odd step per

three steps, so $3\nu_3(t) \leq t + K$ with $K = 3\nu_3(k) + 4$. This equivalence is formalized as `nu3_linear_bound_iff_reaches` in `Conclusion.lean`, confirming that the K-bound is the irreducible core of the Collatz conjecture.

The golden mean shift (Theorem 3.4) makes this plausible. The topological bound $1/\varphi^2$ already lies below p_{eq} by 0.005 (Remark 3.7). Computational evidence strongly supports the hypothesis: over 10^{10} starting values, the observed mean is $p_{\text{odd}} = 0.3324$, well below p_{eq} (Section 8).

Density of obstructions. A natural strategy is to derive (H1) from Baker's theorem via the *structural pure-even sieve* (Definition 8.1, Section 8.3). At each torus scale $(\mathbb{Z}/3^k\mathbb{Z})^2$, certain cells are *structurally* pure-even: every trajectory visiting such a cell is forced to undergo at least two consecutive halvings, not by empirical observation, but by the arithmetic of the 2-adic valuation $v_2(3a_t + 1)$. The Hensel lifting of the $3n+1$ map shows that these cells arise from congruence conditions on $a_t \bmod 2^m$ imposed by the multiplicative identity:

$$(6.1) \quad a_t \cdot 2^{\nu_2(t)} \equiv n \cdot 3^{\nu_3(t)} + C(t) \pmod{3^k}.$$

When ν_2/ν_3 lies in the Baker gap around $\log_2 3$ (i.e., $|\nu_2/\nu_3 - \log_2 3| < C'/(3^k)^{1+\varepsilon}$), these congruence conditions force $a_t \equiv 1 \pmod{4}$, yielding $v_2(3a_t + 1) \geq 2$.

Baker's theorem ensures that the equilibrium line of slope $\log_2 3$ on the torus avoids all rational grid points, so the structural pure-even cells $\mathcal{S}_k$ occupy precisely the Baker gap—the region where the equilibrium line *would* pass if $\log_2 3$ were rational. A trajectory attempting to maintain ν_3/t near p_{eq} must follow the equilibrium line and therefore repeatedly cross $\mathcal{S}_k$ -cells. Each crossing forces extra halvings, pushing ν_3/t below p_{eq} . The Baker–Feldman bridge (Section 8.3) makes this argument precise.

What would suffice to prove (H1)? The remaining gap is a *visitation frequency* bound: one must show that every infinite Collatz trajectory visits $\mathcal{S}_k$ -cells with frequency bounded below by a positive constant independent of n . An ergodic theorem for the Collatz transfer operator, or a spectral gap for the transition matrix on a finite torus quotient, would suffice [11]. Alternatively, one could show directly that the set of parity sequences sustaining $\nu_3/t > p_{\text{eq}} - \delta$ while avoiding $\mathcal{S}_k$ at all scales has Hausdorff dimension < 2 . Tao's result [9] that almost all orbits attain almost bounded values provides partial evidence, though his methods are probabilistic and do not yield the uniform bound needed here.

Discussion of (H2). Baker's theorem [1] is established mathematics. For our purposes, we need the special case $\alpha_1 = 2$, $\alpha_2 = 3$:

$$|m \log 2 - n \log 3| > C / \max(|m|, |n|)^{1+\delta}$$

for effective constants $C, \delta > 0$ depending only on 2 and 3. The gap is not the mathematics but its formalization in Lean 4: the Gel'fond–Schneider proof chain underlying Baker's theorem has no Mathlib precedent (Section 7).

Proof of Theorem B. Assume (H1) and (H2).

Fix $n \geq 1$. By (H1), the walk $w(t)$ has eventually linear drift with some rate $\varepsilon > 0$.

By Theorem A, if the only positive Collatz cycle is $\{1, 2, 4\}$, then $T^k(n) = 1$ for some k .

It remains to verify the cycle hypothesis. For $\Delta_3 = 0$, a cycle is impossible (no odd steps means the trajectory decreases). For $\Delta_3 = 1$, this is Theorem 5.2: the only cycle is $\{1, 2, 4\}$. For $\Delta_3 \geq 2$ with $3\Delta_3 < p$, the strict inequality $a(t) \cdot 2^{\nu_2} < n \cdot 4^{\nu_3}$ contracts over the period. For $\Delta_3 \geq 2$ with $p = 3\Delta_3$ (the balanced case), by (H2), Baker's theorem provides effective lower bounds on $|2^{\Delta_2} - 3^{\Delta_3}|$ that, combined with Steiner-type analysis of the cycle equation, exclude all nontrivial cycles (Remark 5.3). $\square$

Remark 6.1 (Effective bounds). For $\varepsilon = 0.01$ (a modest drift rate): $\alpha = (\log_2 3 + 0.01)/(1 - 0.01) \approx 1.611$, $\rho = 3 \cdot 2^{-1.611} \approx 0.981$, and $B \approx 55$. For $\varepsilon = 0.001$: $B \approx 436$. The bound grows as $B \sim 1/\varepsilon$ for small ε .

7. Lean 4 formalization

The results of this paper have been formalized in Lean 4 using the Mathlib4 library. The formalization comprises approximately 2800 lines across 17 files.

7.1. *What is machine-checked.* The following results are fully proved (zero sorry):

<u>1</u>			
<u>2</u>	Component	File	Key theorem
<u>3</u>			
<u>4</u>	SFT library	<code>SFT.lean</code>	Shift-invariance, word avoidance
<u>5</u>	Golden mean shift	<code>FibCounting.lean</code>	Entropy = $\log \varphi$, $p_{\text{odd}} = 1/\varphi^2$
<u>6</u>	Collatz–SFT bridge	<code>CollatzSFT.lean</code>	Parity sequence in golden mean shift
<u>7</u>	Branch locus	<code>BranchLocus.lean</code>	Cell trichotomy, pure-even forcing
<u>8</u>	Walk infrastructure	<code>Walk.lean</code>	Drift positivity, increment rules
<u>10</u>	Multiplicative independence	<code>Baker.lean</code>	$2^m = 3^n \Rightarrow m = n = 0$
<u>11</u>	Irrationality	<code>Baker.lean</code>	$\log_2 3 \notin \mathbb{Q}$
<u>12</u>	Correction ratio	<code>CorrectionRatio.lean</code>	Recurrence, universal inductive bound
<u>13</u>	Cycle elimination	<code>CorrectionRatio.lean</code>	$\Delta_3 = 0$: contradiction; $\Delta_3 = 1$: $\{1, 2, 4\}$
<u>15</u>			$\Delta_3 \geq 2$: via Baker–Steiner sorry
<u>16</u>	Eventual periodicity	<code>CorrectionRatio.lean</code>	Pigeonhole principle
<u>17</u>	Tunnel width	<code>TunnelWidth.lean</code>	Baker $\rightarrow$ Diophantine gap $\rightarrow$ pure-even
<u>18</u>	Drift infrastructure	<code>Drift.lean</code>	K-bound $\rightarrow \varepsilon$ -gap, walk divergence
<u>19</u>	K-bound equivalence	<code>Drift.lean</code> , <code>Conclusion.lean</code>	K-bound $\Leftrightarrow$ <code>collatzReaches</code>
<u>20</u>	Capstone	<code>Conclusion.lean</code>	<code>collatz_conjecture : CollatzConj</code>
<u>21</u>			

23 7.2. *Remaining sorry axioms.* Seven `sorry` axioms remain, grouped into
24 two categories:
25

<u>26</u>				
<u>27</u>	#	File	Theorem	Category
<u>28</u>				
<u>29</u>	1–5	<code>Baker.lean</code>	<code>baker_aux_construction</code> ,	Gel’fond–Schneider
<u>30</u>			<code>baker_extrapolation</code> ,	
<u>31</u>			<code>baker_zero_estimate</code> ,	
<u>32</u>			<code>baker_effective_bound</code> ,	
<u>33</u>			<code>baker_two_three</code>	
<u>34</u>				
<u>35</u>	6	<code>Baker.lean</code>	<code>baker_no_balanced_cycle</code>	Baker–Steiner cycles ($\Delta_3 \geq 2$)
<u>36</u>	7	<code>Drift.lean</code>	<code>nu3_linear_bound</code>	K-bound (critical path)
<u>37</u>				

38 Items 1–5 form the Baker proof chain: they require formalizing the Gel’fond–
39 Schneider argument for transcendence measures, which has no Mathlib prece-
40 dent. Two former `TunnelWidth.lean` `sorry`s have been resolved: `tunnel_walls_positive_of_baker`
41 was proved by exhibiting cell $(0, 1)$ as structurally even at any scale 3^b (Lemma 8.3);
42

a second sorry on Weyl equidistribution was removed after reformulation (Section 8.3). Item 6 applies Baker’s theorem to exclude non-trivial cycles with $\Delta_3 \geq 2$ and period $p = 3\Delta_3$ (the balanced case), using Steiner-type analysis of the cycle equation combined with computational verification for small Δ_3 . This sorry was previously in `CorrectionRatio.lean` as `no_cycle_equality_case`; it has been restated in pure arithmetic terms in `Baker.lean`, making `CorrectionRatio.lean` sorry-free. Item 7 is the K-bound: the sole sorry on the critical path to `collatz_conjecture`. From the K-bound, the ε -gap (uniform bound $\nu_3/t \leq p_{\text{eq}} - \varepsilon$) is derived as a proved theorem, since $\log_2 3 < 2$ implies $p_{\text{eq}} > 1/3$. The converse is also proved: `collatzReaches(n)` implies the K-bound (via the $1 \rightarrow 4 \rightarrow 2 \rightarrow 1$ cycle), so the K-bound is formally equivalent to the Collatz conjecture (`nu3_linear_bound_iff_reaches`). The Baker–Feldman bridge (Section 8.3) identifies a route to proving this sorry from Baker’s theorem via the structural pure-even sieve; the missing ingredient is a visitation frequency bound.

7.3. *AI/LLM disclosure.* Per Annals policy, we disclose that Claude (Anthropic) was used as a coding assistant for the Lean 4 formalization and for drafting portions of the L^AT_EX manuscript. All mathematical content was developed and verified by the author. The Lean 4 proofs were checked by the Lean kernel, which is independent of any LLM.

8. Computational evidence

The following data are not formal proofs but provide quantitative support for the hypotheses. All computations were performed in C using the program `branch_locus.c`.

8.1. *The 10-billion run.* A complete run over all starting values $n \in [1, 10^{10}]$ was performed (19.5 hours wall time, 2.27×10^{12} total Collatz steps).

Quantity	Value	Note
Starting values N	10^{10}	
Total steps	2.27×10^{12}	
Mean p_{odd}	0.3324	Well below $p_{\text{eq}} \approx 0.387$
“11” violations	0	Over 2.27×10^{12} transitions
Transition fractions	ee = 33.45% eo = 33.16%, oe = 33.39%	

8.2. *Branch locus and tunnel structure.* At torus resolution $k = 144$, the branch locus (cells visited by both even and odd steps) and pure-even walls (cells visited only by even steps) were counted:

k	Branch cells	Pure-even cells	Pure-odd cells
144	16,371	659	0
729	16,577	893	0

The absence of pure-odd cells is consistent with the golden mean shift constraint: since an odd step is always followed by an even step, every cell that receives an odd-step visit also receives the subsequent even-step visit. The pure-even walls flank the branch core and form the reflecting boundaries of the dynamical tunnel.

8.3. Structural pure-even cells and the Baker–Feldman bridge. The definition of pure-even cells used by the computational infrastructure (Section 8) is *empirical*: a cell is “pure-even” if no trajectory in a finite sample visited it with an odd step. This conflates an observation about specific trajectories with a property of the cell itself. We now give a *geometric* definition that depends only on the arithmetic of the cell, not on which trajectories happen to visit it.

Definition 8.1 (Structural pure-even cell). A cell $(a, b) \in (\mathbb{Z}/3^k\mathbb{Z})^2$ is *structurally pure-even* if every $n \geq 1$ that visits (a, b) —i.e., for which $(\nu_2(n, t) \bmod 3^k, \nu_3(n, t) \bmod 3^k) = (a, b)$ at some step t —has *even* iterate a_t at that step. In other words, every visit to the cell occurs at an even step (so the walk increment is $+1$), and no trajectory can take an odd step while occupying this cell.

Definition 8.2 (Forces double halving). A cell $(a, b) \in (\mathbb{Z}/3^k\mathbb{Z})^2$ *forces double halving* if every visit with an odd iterate a_t satisfies $v_2(3a_t + 1) \geq 2$, so that at least two consecutive halvings follow the odd step. Since structural pure-even cells have no odd visits, they force double halving vacuously.

The condition $v_2(3m + 1) \geq 2$ is equivalent to $m \equiv 1 \pmod{4}$: one checks $3 \cdot 1 + 1 = 4$, and in general $3m + 1 \equiv 4 \pmod{8}$ when $m \equiv 1 \pmod{4}$. This characterizes cells that *force double halving* (Definition 8.2): after an odd step at such an iterate, the trajectory divides by at least 4 before the next odd step, producing two even steps where only one was guaranteed. Structural pure-even cells (Definition 8.1) are a distinct, stronger property: the iterate is forced to be *even* by the torus arithmetic, so the step is even and no odd-step analysis applies.

Hensel lifting of the $3n + 1$ map. The Collatz map lifts naturally to the 2-adic integers $\mathbb{Z}_2$. Given $n \in \mathbb{Z}_2$ with n odd, the map $n \mapsto (3n + 1)/2^{v_2(3n+1)}$ is well-defined and continuous in the 2-adic topology. The residue class $n \bmod 2^m$ determines the first $\sim m$ parity steps of the trajectory: if $n \equiv r \pmod{2^m}$,

then $v_2(3n+1) = v_2(3r+1)$, and the parity of the first $v_2(3r+1)$ iterates is completely determined.

At torus scale 3^k , the multiplicative identity (2.2) constrains $n \bmod 3^k$ and $n \bmod 2^m$ simultaneously. A cell (a, b) on $(\mathbb{Z}/3^k\mathbb{Z})^2$ forces $\nu_3 \equiv b \pmod{3^k}$, which via the identity imposes a congruence on $a_t \bmod 2^m$ for $m \asymp k \log_2 3$. When this forced residue class satisfies $a_t \equiv 0 \pmod{2}$, the cell is structurally pure-even: every trajectory visiting it takes an even step, regardless of the starting value. When the forced class satisfies $a_t \equiv 1 \pmod{4}$ (necessarily an odd iterate), the cell forces double halving (Definition 8.2).

The sieve. At each scale 3^k , the structurally pure-even cells form a subset $\mathcal{S}_k \subset (\mathbb{Z}/3^k\mathbb{Z})^2$. As k increases, the Chinese Remainder Theorem interleaves the 2-adic and 3-adic constraints, and the fraction $|\mathcal{S}_k|/3^{2k}$ stabilizes at a positive value. The cells in $\mathcal{S}_k$ act as a sieve: any trajectory entering one is forced to take an even step (walk increment $+1$).

LEMMA 8.3 (Forced even descent). *For each $k \geq 1$, let $\mathcal{S}_k$ denote the set of structurally pure-even cells in $(\mathbb{Z}/3^k\mathbb{Z})^2$ (Definition 8.1). Then:*

- (i) $\mathcal{S}_k \neq \emptyset$ for all $k \geq 1$.
- (ii) For any $n \geq 1$ and any step t such that $(\nu_2(n, t) \bmod 3^k, \nu_3(n, t) \bmod 3^k) \in \mathcal{S}_k$, the iterate a_t is even, so the step is an even step and the walk increment is $+1$.
- (iii) $|\mathcal{S}_k|/3^{2k} \geq c > 0$ for a constant c independent of k .

Evidence. Part (i) is proved in Lean 4: cell $(0, 1)$ is structurally pure-even at every scale 3^b (`tunnel_walls_positive_of_baker`). Part (ii) follows from the definition. Part (iii) is verified computationally for $k \leq 4$: at scale 3^2 (81 cells), 19 cells remain structurally pure-even after examining trajectories 1 through 10 over 50 steps; the count grows with k .

The Baker–Feldman bridge. The connection between Baker’s theorem and the K-bound (`nu3_linear_bound`) proceeds in three steps.

- (1) *Baker constrains the equilibrium line.* On the torus $(\mathbb{Z}/3^k\mathbb{Z})^2$, the equilibrium line has slope $\log_2 3$. Baker’s lower bound

$$|m \log 2 - n \log 3| > C / \max(|m|, |n|)^{1+\delta}$$

implies that this line maintains distance $\gg (3^k)^{-\delta}$ from every rational grid point (p, q) with $0 \leq p, q < 3^k$. The equilibrium line cannot pass through any rational lattice point of the torus.

- (2) *Sieve cells and the Baker gap.* The structural pure-even cells $\mathcal{S}_k$ are located where the multiplicative identity (2.2) constrains the iterate to

be even. Baker’s theorem ensures these cells cluster near the equilibrium line but never on it, creating a “sieve belt” of forced even steps around the irrational slope $\log_2 3$.

- (3) *The trajectory cannot ride the equilibrium.* Under equidistribution of the winding-number path on the torus, the trajectory visits $\mathcal{S}_k$ -cells with positive frequency $\delta > 0$. Each such visit forces an even step (walk increment $+1$). Combined with the heuristic that the average 2-adic valuation $v_2(3a_t + 1)$ per odd step is $E[v_2] = 2 > \log_2 3 \approx 1.585$ (from near-uniform distribution of $a_t \bmod 2^m$), the walk has positive drift rate. The correction ratio bound (Theorem 4.4) then gives eventual boundedness, periodicity, and the K-bound via cycle elimination.

The rigorous version requires proving that the average v_2 per odd step exceeds $\log_2 3$ for every trajectory—not just under the random model.

Specifically, the following would suffice to close the gap:

- (1) *Average halving bound:* For every $n \geq 1$, the average 2-adic valuation per odd step satisfies $\liminf_{k \rightarrow \infty} v_2(\tau_k)/k > \log_2 3$, where $\tau_1 < \tau_2 < \dots$ are the odd-step times. Equivalently, $v_3(t)/t < p_{\text{eq}}$ eventually. Under the random model, $E[v_2(3a + 1)] = 2$ for odd a uniformly distributed, giving margin $2 - \log_2 3 \approx 0.415$.
- (2) *Equidistribution:* A uniform distribution result for $a_t \bmod 2^m$ (conditional on a_t being odd) at a fixed depth $m \geq 2$. Even $m = 2$ suffices: if $a_t \bmod 4$ is uniformly distributed among $\{1, 3\}$ at odd steps, then half have $v_2 = 1$ and half have $v_2 \geq 2$, giving $E[v_2] \geq 3/2$. Combined with the contribution from $v_2 \geq 3$ cases, this exceeds $\log_2 3$.

Tao’s result that almost all orbits attain almost bounded values provides partial evidence, though his methods are probabilistic and do not yield the uniform bound needed here.

By the K-bound equivalence (`nu3_linear_bound_iff_reaches` in `Conclusion.lean`), the K-bound is formally equivalent to the Collatz conjecture for each n , so any route to the K-bound—whether through the Baker–Feldman bridge or otherwise—would resolve the conjecture.

Remark 8.4 (Computational results are not proofs). The computational data of Section 8 verify the golden mean shift constraint and the tunnel structure for $N = 10^{10}$, but they do not constitute formal proofs. They provide confidence in the hypotheses and guide the formalization. The observed $p_{\text{odd}} = 0.3324$ is a mean over all trajectories; individual trajectories may temporarily exceed p_{eq} . The Baker–Feldman bridge above identifies the precise mathematical content needed to close the gap.

Acknowledgements. The author thanks the Lean 4 and Mathlib communities for the proof assistant infrastructure that made the formalization possible.

References

1. A. Baker, *Transcendental Number Theory*, Cambridge University Press, 1975. [doi:10.1017/CB09780511565977](https://doi.org/10.1017/CB09780511565977).
2. J. H. Conway, Unpredictable iterations, *Proc. Number Theory Conf. (Boulder, CO, 1972)*, pp. 49–52, Univ. Colorado, Boulder, 1972.
3. S. Eliahou, The $3x + 1$ problem: new lower bounds on nontrivial cycle lengths, *Discrete Math.* **118** (1993), 45–56. [doi:10.1016/0012-365X\(93\)90052-U](https://doi.org/10.1016/0012-365X(93)90052-U).
4. J. C. Lagarias, The $3x + 1$ problem and its generalizations, *Amer. Math. Monthly* **92** (1985), no. 1, 3–23. [doi:10.1080/00029890.1985.11971528](https://doi.org/10.1080/00029890.1985.11971528).
5. J. C. Lagarias (ed.), *The Ultimate Challenge: The $3x + 1$ Problem*, Amer. Math. Soc., Providence, RI, 2010. [doi:10.1090/mbk/078](https://doi.org/10.1090/mbk/078).
6. T. Oliveira e Silva, S. Herzog, and S. Pardi, Empirical verification of the $3x + 1$ and related conjectures, *The Ultimate Challenge: The $3x + 1$ Problem* (J. C. Lagarias, ed.), Amer. Math. Soc., 2010, pp. 189–207.
7. J. Simons and B. M. M. de Weger, Theoretical and computational bounds for m -cycles of the $3n + 1$ problem, *Acta Arith.* **117** (2005), no. 1, 51–70. [doi:10.4064/aa117-1-3](https://doi.org/10.4064/aa117-1-3).
8. R. P. Steiner, A theorem on the Syracuse problem, *Congr. Numer.* **20** (1977), 553–559.
9. T. Tao, Almost all orbits of the Collatz map attain almost bounded values, *Forum Math. Pi* **10** (2022), Paper No. e12, 56 pp. [doi:10.1017/fmp.2022.8](https://doi.org/10.1017/fmp.2022.8).
10. R. Terras, A stopping time problem on the positive integers, *Acta Arith.* **30** (1976), no. 3, 241–252. [doi:10.4064/aa-30-3-241-252](https://doi.org/10.4064/aa-30-3-241-252).
11. G. J. Wirsching, *The Dynamical System Generated by the $3n + 1$ Function*, Lecture Notes in Math., vol. 1681, Springer-Verlag, Berlin, 1998. [doi:10.1007/BFb0095985](https://doi.org/10.1007/BFb0095985).

JOHN.JANIK@GMAIL.COM